HENG ZENG XU (JASON)

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PROFESSIONAL SUMMARY

Computer Science graduate from Monash University Malaysia (coursework completed; convocation November 2026) with a focus on backend architecture, resilient system design, and AI integrations. A fast, self-driven learner who continuously builds real-world applications, audits edge cases, and leverages modern AI-assisted engineering workflows. Eager to bring strong technical initiative and problem-solving skills to a full-time software engineering role.

PROJECTS

Precisely – AI-Native ERP & WhatsApp Ordering Assistant 2026

Next.js · TypeScript · Supabase / PostgreSQL · Anthropic API · WhatsApp Cloud API | github.com/xengzuh/precisely

- Designed and built the entire system solo, spanning a 22-table PostgreSQL database, 31 pages/APIs, and ~19,000 lines of TypeScript using Next.js, Supabase, and Claude AI.
- Automated WhatsApp order-taking by linking Meta's WhatsApp Cloud API to an AI agent querying live stock levels, achieving end-to-end response and order insertion in ~10 seconds.
- Enforced strict risk classification to restrict auto-replies to 3 of 8 inquiry types, deterministically routing pricing, delivery schedules, and safety inquiries to human staff.
- Engineered a deterministic unit conversion engine covered by 135 automated unit tests to eliminate volume-to-weight calculation errors (e.g., liters vs. kilograms).
- Audited background jobs running outside standard session auth, resolving cross-tenant data exposure risks across 5 critical database operations.
- Routed both AI operations and web interface interactions through a single, logged execution path across all 25 business operations to ensure auditability and reversibility.

CloudBoard – Smart Digital Bulletin Board (Final Year Project) 2026

Next.js 15 · React 19 · Supabase · TensorFlow.js · Google Cloud Vision

- Architected real-time digital signage software that synchronizes content across connected displays instantly, featuring a custom drag-and-drop canvas layout editor.
- Integrated on-device person detection with TensorFlow.js to trigger context-aware content, paired with automated media moderation via Google Cloud Vision and Video Intelligence.

Santorini – Desktop Strategy Board Game 2025

Python · Tkinter · Pillow · Object-Oriented Design | github.com/xengzuh/Santorini

- Engineered a standalone desktop board game in Python featuring full rule enforcement, configurable god powers, match timers, and a standalone executable packaged for Windows.
- Decoupled game engine logic entirely from presentation layers, utilizing an extensible shared abstraction for rule-altering player powers and custom canvas primitives for animations without third-party game engines.

EXPERIENCE

Web Developer Intern – Impreszions Solutions Dec 2025 – Feb 2026

- Engineered and deployed a production corporate website for law firm **Law Kuan Yew & Co.** (lkyco.com.my), delivering responsive UI layouts and customized service landing pages to specification.
- Collaborated in an agile team to prototype an interactive, museum-based educational treasure hunt mobile web app, leveraging rapid AI-assisted development workflows ("vibe coding") to build core gamified learning features.

EDUCATION

Bachelor of Computer Science – Monash University Malaysia Coursework Completed (Convocation Nov 2026)

Relevant Coursework: Data Structures, Algorithms, Database Systems, Operating Systems, Mobile App Development, Parallel Computing

TECHNICAL SKILLS

Languages:	TypeScript, JavaScript, Python, Java, SQL, HTML/CSS
Backend & Architecture:	PostgreSQL, Supabase, Node.js, REST APIs, Webhooks, Relational Schema Design, Row-Level Security (RLS)
Frontend & Full Stack:	Next.js (App Router), React 19, Tailwind CSS, WordPress
AI, Cloud & Tools:	Anthropic API (Claude), Prompt Engineering & Guardrails, TensorFlow.js, Google Cloud Platform, Git, Automated Testing